

Ishan Chhaparwal

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SUMMARY

Results-driven AI/ML Developer skilled in Machine Learning, Deep Learning, NLP, Generative AI, and Agentic AI, with hands-on experience building and deploying multiple production-grade AI systems — including multi-agent architectures, RAG pipelines, and recommendation engines — using Python, TensorFlow, Scikit-learn, LangChain, LangGraph, and Hugging Face Transformers. Author of a published AI research paper and filed patent in Machine Learning, with 450+ Data Structures & Algorithms problems solved.

EDUCATION

B.Tech in Information Technology — Vishwakarma Institute of Information Technology, Pune

2023 – 2027

- CGPA: 8.99 / 10.0

TECHNICAL SKILLS

Languages: Python, C++, SQL

Machine Learning: Deep Learning, Supervised & Unsupervised Learning, TensorFlow, Scikit-learn, Feature Engineering, Data Preprocessing, Model Evaluation, Recommendation Systems

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Generative AI / NLP / Agentic AI: LLMs, LangChain, LangGraph, Hugging Face Transformers, RAG, ChromaDB, Vector Embeddings, Semantic Search, Whisper

Database & Deployment: MySQL, MongoDB, Streamlit, Render, Vercel

Tools & Core CS: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Data Structures & Algorithms (450+ Problems), OOP, DBMS, Operating Systems

PUBLICATION & PATENT

- **Research Publication: "NeuroSync: Monitoring and Analysis of Psychological Impact of Prolonged AI Interactions on Humans"** — proposed an AI-driven framework using Machine Learning, NLP, Sentiment Analysis, and Behavioral Analytics to monitor the psychological impact of prolonged human–AI interaction, supporting ethical AI and mental well-being assessment.
- **Patent Filed:** "Hybrid ML-NLP Resume Analyzer and Intelligent Job Recommendation System with Cloud-Native MLOps Architecture" — an AI-powered recruitment automation system integrating Machine Learning, NLP, recommendation systems, and cloud-native MLOps for intelligent resume analysis and personalized job recommendations.

PROJECTS

TripMate AI — Multi-Agent Travel Planning Assistant

Jul 2026 – Aug 2026

GitHub: [Ishan-1104/TripMate-AI](#) | Live Demo: [tripmate-a.streamlit.app](#)

Python • LangGraph • LangChain • Groq • Duffel API • AviationStack • Tavily • PostgreSQL • Streamlit

- Architected a multi-agent AI travel planning system using **LangGraph** and **LangChain**, orchestrating **4+** specialized agents (flight, hotel, itinerary, budget) that reduced end-to-end trip-planning time from **~2 hours of manual research to under 10 minutes** via a single natural-language query.
- Integrated **Duffel API**, **AviationStack**, and **Tavily Search** with **PostgreSQL**-backed checkpointing to deliver real-time flight/hotel data and persistent itineraries, supporting **50+ test queries** across **50+ cities** with **95%** query success rate on the deployed Streamlit app.

AI Video Assistant — RAG-Powered Meeting & Video Intelligence Platform

Jun 2026 – Jul 2026

GitHub: [Ishan-1104/Video-Agent](#) | Live Demo: [video-agent-18.streamlit.app](#)

Python • LangChain (LCEL) • Mistral AI • Whisper • ChromaDB • Hugging Face • Streamlit

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain LCEL**, **Mistral AI**, and **Whisper** to process YouTube/local videos, auto-generating transcripts, summaries, action items, and key decisions, with a session-isolated semantic search system (**ChromaDB + all-MiniLM-L6-v2**) enabling conversational Q&A across **1,000+ transcript chunks** with **zero cross-session data leakage**.
- Engineered and shipped a production-ready Streamlit application supporting **concurrent user sessions**, real-time processing status, transcript analytics, session history, and an interactive AI chat assistant.

CineAI — AI-Powered Movie Recommendation Platform

Apr 2026 – May 2026

GitHub: [Ishan-1104/CineAI](#) | Live Demo: [cineai-6ygv.onrender.com](#)

Python • Scikit-learn • TF-IDF • Cosine Similarity • React.js • Express.js • MongoDB • TMDB API

- Built a content-based movie recommendation engine using **TF-IDF Vectorization** and **Cosine Similarity** across **500,000+ TMDB movie records**, integrating **5+ TMDB APIs** for real-time search, metadata, and poster retrieval, and delivering **sub-500ms response times** at scale.
- Developed and launched a full-stack platform (**React.js**, **Express.js**, **MongoDB**, **JWT authentication**) connecting the recommendation engine to a real-time backend, enabling secure accounts, personalized watchlists, and favorites management with **99.9% uptime**.

ACHIEVEMENTS

- **Competitive Programming: 450+ DSA Problems Solved (LeetCode)** — spanning Arrays, Strings, Linked Lists, Trees, Graphs, Dynamic Programming, Greedy Algorithms, Binary Search, and Backtracking.
- **SEC Club - Operations Domain (Member)**

2023-2024